

AI-Driven Loan Default Prediction Using Banking Analytics

Explainable Machine Learning for Credit Risk and Default Prediction in Banking

1. Refined Title

AI-Driven Loan Default Prediction Using Banking Analytics: Explainable Machine Learning for Credit Risk Assessment

The title reflects the two core threads carried through the project: banking-analytics-driven loan default prediction, and explainability as a first-class requirement rather than an afterthought — a distinction the literature survey below shows is still uncommon.

2. Problem Statement

Manual underwriting and legacy scorecards cannot capture the complex, non-linear relationships between borrower behaviour, repayment history, and macroeconomic conditions. Traditional models routinely misclassify risk, directly inflating non-performing assets (NPAs) and reducing capital efficiency for banks and NBFCs. Where institutions do adopt machine learning, most models offer no explanation for a decision, making them hard to trust, audit, or defend to a regulator. Default cases are also a small minority of records, and most studies validate on a single dataset, limiting how well findings generalise across lending contexts.

3. Literature Survey

Eleven recent studies (2024–2026) from IEEE, ACM, and Springer venues were reviewed:

Year	Author(s)	Paper	Method / Finding	Gap / Relevance
2024	Aruleba, I., & Sun, Y.	Effective Credit Risk Prediction Using Ensemble Classifiers With Model Explanation	Ensemble classifiers (Random Forest, AdaBoost, XGBoost, LightGBM) combined with SMOTE-ENN resampling and SHAP explanations on German and Australian credit datasets.	Strong ensemble + SHAP result, but validated on only two credit datasets — limited cross-context testing.
2024	Zhu, M., Zhang, Y., Gong, Y., et al.	Ensemble Methodology: Innovations in Credit Default Prediction Using LightGBM, XGBoost, and LocalEnsemble	LightGBM–XGBoost–local-ensemble stacking for credit default prediction; ensembling lifts detection over single learners.	Shows stacking gains over single learners, but includes no explainability layer for the final ensemble.
2024	Vihurskyi, B.	Credit Card Fraud Detection with XAI: Improving Interpretability and	Explainable AI applied to credit-card fraud/default detection to improve interpretability and	Builds analyst trust via XAI, but scope is limited to one dataset and fraud-flagging rather than a full credit-risk pipeline.

		Trust	analyst trust.	
2025	Gonaygunta, H., Maturi, M. H., Yadulla, A. R., et al.	Utilizing Explainable AI in Financial Risk Assessment: Enhancing User Empowerment through Interpretable Credit Scoring Models	Interpretable credit-scoring model using explainable AI to empower end users in financial risk assessment.	Empowers end users with interpretable scoring, but XAI is applied to a single chosen model, not compared across candidates.
2025	Hartomo, K. D., Arthur, C., & Nataliani, Y.	A Novel Weighted Loss TabTransformer Integrating Explainable AI for Imbalanced Credit Risk Datasets	Weighted-loss TabTransformer integrating explainability for imbalanced credit-risk datasets.	Handles class imbalance well with built-in explainability, but transformer complexity trades off transparency; tested on one dataset.
2025	ACM GAIIS '25 (author not specified in listing)	Loan Default Prediction Based on Machine Learning Approaches	Seven-model comparison (Decision Tree, Random Forest, AdaBoost, Bagging, XGBoost, LightGBM, CatBoost) on Alibaba Tianchi loan data; CatBoost reported best on AUC (0.7488).	Broad model comparison, but restricted to one dataset with no explainability component.
2025	ACM CSAIDE '25 (author not specified in listing)	Using Explainable Machine Learning to Predict Loan Risk in Consumer Finance	Explainable ML with SHAP on the Home Credit dataset (1.53M records, 3.14% default rate) for consumer-finance risk.	Explainability improves user trust, but SHAP is applied only after model selection, not during model comparison.
2024	Demma Wube, H., Zekarias Esubalew, S., Fayiso Weldesellase, F., & Girma Debelee, T.	Deep Learning and Machine Learning Techniques for Credit Scoring: A Review	Systematic review of deep-learning vs. classical ML techniques for credit scoring, PanAfriConAI 2023.	Confirms the accuracy–interpretability trade-off across the field, but is a review rather than an implemented framework.
2026	Monje, L., Carrasco, R. A., & Sánchez-Montaños, M.	Machine Learning XAI for Early Loan Default Prediction	XGBoost with post-hoc SHAP explanations for early loan-default prediction on P2P lending data.	Strong early-warning result, but validated on a single P2P lending dataset only.
2025	Liang, K., Ye, C., & Zhao, J.	Research on Credit Risk Analysis of Technology-based Small and Microenterprises Based on Explainable Deep Learning	Explainable deep learning (LSTM with attention) for credit-risk analysis of technology-based small and micro enterprises.	Domain-specific explainable DL for one lending context (TSMEs); findings don't generalize to mainstream retail lending.
2026	Kofidis, K., Cocianu, C. L., & Uscatu, C.	Dealing with Imbalanced Data for Loan Default Prediction: A Comparative Study Across Balancing Techniques for CNN, LSTM, RF, and SVM Models	Comparative study of balancing techniques (SMOTE family) across CNN, LSTM, Random Forest, and SVM for loan default.	Thorough comparison of balancing techniques across four model families, but no explainability or decision-support layer integrated.

Literature synthesis: strong single-model or single-dataset accuracy gains, but explainability is usually bolted on to one “winning” model rather than compared across models, datasets, or lending contexts.

4. Research Gaps Identified

- Single-dataset validation – Papers 1, 3, 5, 6, 9, 10 each validate on one dataset (German credit, Home Credit, or a single P2P platform) — findings rarely transfer across lending environments.
- Explainability applied late – SHAP/LIME (papers 1, 3, 5, 6, 7, 9) is typically bolted on to a single “winning” model after selection, not used to compare candidate models during development.
- Accuracy–interpretability trade-off – Deep/transformer models (papers 5, 10) push accuracy up, but interpretability studies (paper 8) show this often comes at the cost of transparency for non-technical stakeholders.
- No decision-support layer – Almost none of the surveyed work (papers 1–11) packages results into a dashboard loan officers can actually use — most stop at reporting Accuracy/F1/AUC tables.

5. Objectives

- Consolidate benchmark data – Integrate Lending Club, Statlog German Credit, and Home Credit datasets into one structured, analysis-ready pipeline using schema alignment and unified feature encoding.
- Perform deep, decision-grade EDA – Run 12+ analytical techniques — missingness, outliers, class-imbalance, and correlation analysis — before any modelling, using univariate and multivariate statistical methods.
- Engineer risk-aware features – Derive debt-to-income ratio, credit-utilization, and repayment-consistency features using domain-driven feature engineering to sharpen predictive signal.
- Benchmark five ML models – Train and compare Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost under identical, cross-validated conditions.
- Make predictions explainable – Apply SHAP and LIME to the best model so every default-risk score has an auditable, human-readable reason.
- Deliver a usable dashboard – Package results into a visualization dashboard so loan officers see the prediction and the reasoning together.

6. Novelty / Innovation, Feasibility, and Project Plan

6.1 Novelty / Innovation

Each identified gap maps to a specific design decision that is new relative to the surveyed work:

- Single-dataset validation → Three benchmark datasets combined — Lending Club, Statlog German Credit, Home Credit — spanning traditional, classical, and thin-file/alternative lending.
- Explainability applied late → SHAP and LIME run across all five candidate models during comparison, not only on the final chosen model.

- Accuracy vs. interpretability trade-off → Five models spanning simple-to-complex (Logistic Regression → CatBoost) benchmarked on the same metrics, letting stakeholders choose their own point on the trade-off curve.
- No decision-support layer → A visualization dashboard (heatmaps, boxplots, feature-importance charts) turns model output into something a loan officer can act on, not just a metrics table.

6.2 Feasibility

- Data feasibility – 3 datasets are public, well-documented, and widely cited in credit-risk research; combined size supports robust train/test/validation splits.
- Technical feasibility – All five ML models have mature, well-supported Python libraries; SHAP and LIME are standard, actively maintained explainability toolkits; class imbalance is addressable (SMOTE, class-weighting).
- Economic feasibility – Built entirely on open-source tools and free/public datasets — no licensing cost.
- Time feasibility – Scoped to fit the semester timeline with buffer before final submission; a modular pipeline lets EDA, modelling, and XAI stages proceed in parallel.

6.3 Project Plan

Phase	Milestone	Key Activities
1	Data Acquisition & Cleaning	Collect and merge 3 datasets, handle missing values, duplicates, encoding.
2	Exploratory Data Analysis	12+ techniques: univariate through multivariate, correlation, imbalance analysis.
3	Feature Engineering & Selection	DTI ratio, credit utilization, mutual information, RFE.
4	Model Building & Comparison	Train LR, Decision Tree, Random Forest, XGBoost, CatBoost.
5	Explainability & Evaluation	SHAP + LIME analysis; Accuracy/Precision/Recall/F1/ROC-AUC.
6	Dashboard & Final Reporting	Build visualization dashboard; consolidate findings for submission.

7. Team Details

Role	Name
Team Member	Tejasree Sagiraju
Team Member	Sruthi Bhatraju
Team Member	B. Bhavya Sai Sree
Guide	K. Swapnika
Course	Data Analytics & Visualisation – 1 (24DSB3101), A.Y. 2026–2027
Review	Review-1 Presentation · 5 August 2026

8. References / Source Links

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